```
import java.util.Arrays;
import java.util.Random;

public class BubbleSort {

	// Sorting method
	public static void sort(int[] arr) {

		for (int i = 0; i < arr.length - 1; i++) {
			for (int j = 0; j < arr.length - 1 - i; j++) {

				if (arr[j] > arr[j + 1]) {
					int temp = arr[j];
					arr[j] = arr[j + 1];
					arr[j + 1] = temp;
				}
			}
		}
	}

	// Run this algorithm individually
	public static void main(String[] args) {

		int[] arr = new int[1000];
		Random random = new Random();

		for (int i = 0; i < arr.length; i++) {
			arr[i] = random.nextInt(10000);
		}

		System.out.println("Before Sorting:");
		System.out.println(Arrays.toString(arr));

		long startTime = System.nanoTime();
		sort(arr);
		long endTime = System.nanoTime();

		System.out.println("After Bubble Sort:");
		System.out.println(Arrays.toString(arr));

		System.out.println("Execution Time: "
				+ (endTime - startTime) + " nanoseconds");
	}
}
```

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